

ADITEYA VARMA KALIDINDI

Portfolio | aditeya.varma@gmail.com | LinkedIn | Github | Los Angeles, CA

EXPERIENCE

Amazon, Software Development Engineer Intern | San Diego, CA

May 2025 – Aug 2025

- Accelerated data processing as measured by a **22% reduction in ETL pipeline latency**, by refactoring Scala-based query generation templates for Neo4j.
- Intercepted **98% of performance regressions** prior to release with **zero metric ingestion loss** by architecting an end-to-end telemetry pipeline using AWS SQS and CloudWatch, coupled with a serverless AWS EMR and DynamoDB log analysis engine.
- Streamlined integration testing from **multi-day feedback loops to sub-15-minute automated checks**, by developing a scalable testing framework using AWS CDK and TypeScript.

Home Centre (Landmark Group), Data Engineer Intern | Bengaluru, India

Jan 2024 – May 2024

- Scaled competitor pricing intelligence as measured by **99.5% data consistency across 1,000+ SKUs**, by building a JavaScript web-scraping engine featuring custom retry logic and proxy rotation.
- Drove category revenue growth as measured by a **7.6% uplift**, by analyzing **500k+ weekly transactions across 95+ locations** to identify attribute-based purchasing correlations.
- Simplified inventory operations as measured by the consolidation of **95 individual store pipelines into 4 unified streams**, by implementing an attribute-based clustering framework.

Open Source Contributor: sageox/ox | Go, SQLite, Git Internals, Concurrency

- Fixed a race condition in **3 SQLite migration functions** that crashed the database on concurrent, cold-start initialization, by engineering a two-phase goroutine-barrier test harness that forced the exact microsecond-wide collision on demand.
- Stopped a data-corruption bug across **2 separate auto-commit paths** that had already baked broken conflict markers into **8 sessions' permanent history over 3 months**, by guarding every commit against staged conflicts using git's index instead of the working tree.
- Killed a fork bomb in the test suite that hung for **10 minutes**, cutting runtime by **66%** once fixed, by stopping the session-prime code from re-executing the compiled test binary against itself.

TECHNICAL SKILLS

Languages: Python, Go, TypeScript, JavaScript, Scala, C, SQL, Bash

AI & LLM Systems: LLM APIs, AI Agents (LangGraph), Multi-Agent Systems, RAG, Prompt Engineering, Evaluation (Ragas)

Developer Tools: Claude Code, Cursor, Git, GitHub Actions, CI/CD, Docker, Kubernetes, Linux

Backend & Data: REST APIs, FastAPI, React, ETL Pipelines, Apache Kafka, Flink, Redis, PostgreSQL, DynamoDB

Cloud, Observability & Testing: AWS (S3, SQS, EMR, CDK, CloudWatch), OpenTelemetry, Prometheus, Grafana, PyTest

TECHNICAL PROJECTS

Darwin Server Runtime: Native Process Isolation | Swift, C, launchd, Mach APIs, OpenTelemetry

- Decreased container overhead and startup latency as measured by a **55.4 ms cold-spawn (2.3x faster** than Colima) and a **2.9 MB idle footprint (160x smaller** than Docker), by engineering a native Darwin process isolation runtime in **Swift and C**.
- Achieved real-time observability for isolated workloads by architecting a telemetry daemon that polled Mach APIs, buffered metrics via an actor-guarded queue, and exported traces via OTLP/HTTP to stream process memory and CPU throughput.

Janus 2.0: Agentic AI Regulation System | Python, LangGraph, Redis, React 19, Ragas

- Resolved reasoning conflicts across FIA technical frameworks as measured by a **90% Answer Relevancy** score on **20-query evaluation benchmarks**, by architecting a metadata-filtered retrieval pipeline and automating tests via the **Ragas framework**.
- Enabled durable, long-term multi-agent reasoning as measured by accurate thread-isolated session memory across distributed users, by engineering a stateful agentic system using **LangGraph** and an external **Redis (RedisSaver)** persistence layer.

Media Partner Ingestion Validator | Python, FastAPI, Docker, TypeScript, PyTest

- Cut manual partner-feed review errors to **0**, by building dual-schema validators for metadata and **HLS/DASH manifests**.
- Delivered deterministic defect detection prior to partner onboarding as measured by a **100% pass rate (22/22) on HTTP 404 segment failure paths**, by authoring a network-free integration test suite utilizing **in-process ASGI transport**.

Multi-Threaded Traffic Analyzer & Caching DNS Proxy | Python, POSIX Sockets, Scapy, PyTest

- Slashed repeat domain resolution latency **from 45ms+ to sub-1ms** for cached hits, by implementing a zero-dependency **RFC 1035 DNS wire-parser** featuring a thread-safe LRU/TTL cache that eliminated redundant upstream forwarding to 8.8.8.8.
- Ensured proxy resilience against network anomalies as measured by successfully passing a **33-test deterministic suite**, by authoring automated fault-injection pipelines using **PyTest and Scapy** to simulate socket exhaustion and malformed headers.

Voyager: Distributed Video Processing Engine | Kafka, Flink, S3, Redis, Python

- Powered real-time stream processing for multi-camera video feeds as measured by sustaining a **sub-50ms per-frame latency** target across decode and inference stages, by building a distributed pipeline using a **6-partition Apache Kafka topic and PyFlink**.
- Curbed heavy neural network compute load as measured by a **75%+ cache hit rate** that proportionally eliminated redundant MobileNetV2 inference calls, by engineering a perceptual embedding cache using **dHash and Redis**.

EDUCATION

University of Southern California, Master of Science in Computer Science (GPA: 3.6/4.0) | Los Angeles, CA

May 2026

Ramrao Adik Institute of Technology, Bachelor of Technology in Computer Engineering (GPA: 9.64/10) | Mumbai, India

May 2024